



AZURE PROJECT REFINEMENT REPORT

AZURE INFRASTRUCTURE DEPLOYMENT AND SECURITY ENHANCEMENT

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EXECUTIVE SUMMARY

The purpose of this project was to design and implement a secure cloud infrastructure using Microsoft Azure for a fictional organization, Ensign Technologies. The project required creating organizational resource groups, configuring Microsoft Entra ID users and security groups, provisioning Azure Storage resources, assigning role-based access permissions, and preparing the environment for future virtual machine deployment.

Throughout the project, I strengthened my understanding of cloud administration while gaining practical experience with Azure governance, identity management, storage security, and access control. I also learned that successful cloud deployments require careful planning, continuous testing, and ethical decision-making to protect organizational data.

After evaluating my original implementation, I identified several areas for improvement, including permission management, security practices, documentation, and deployment planning. These refinements increased both the security and reliability of the environment.

ORIGINAL PROJECT OBJECTIVES

The project was designed to accomplish the following objectives:

- ☐ Create Azure Resource Groups to organize company resources.
- ☐ Configure Microsoft Entra ID users and departmental groups.
- ☐ Create Azure Storage resources for multiple departments.
- ☐ Allocate storage resources for Sales and Marketing, Engineering, and Customer Success.
- ☐ Restrict access using Azure Role-Based Access Control (RBAC).
- ☐ Prepare infrastructure for future virtual machine deployment.
- ☐ Demonstrate secure cloud administration practices.

These objectives reflected many of the responsibilities performed

TECHNICAL CHALLENGES ENCOUNTERED

During implementation, I encountered several technical challenges that required research, troubleshooting, and persistence.

Challenge 1: Azure Subscription and Sandbox Access

Initially, I experienced difficulty accessing the required Azure subscription because my personal Azure account did not have an active subscription in the default directory. This prevented me from creating resources until I verified my student account and gained access to the appropriate environment.

Resolution

I worked through the account verification process, selected the correct Azure directory, and ensured that I was operating within the proper subscription before continuing with the project.

Challenge 2: Azure Policy Restrictions

When creating Azure resources, deployment validation failed because organizational policies restricted resource creation in certain regions.

Resolution

After reviewing the deployment error, I selected an approved Azure region that complied with the organization's policy requirements. This experience taught me how Azure Policy enforces governance and organizational standards.

Challenge 3: Understanding Azure Storage Options

Choosing among different storage redundancy options, managed disks, and performance tiers was initially confusing.

Resolution

I researched Azure storage redundancy models, including Locally Redundant Storage (LRS), Geo-Redundant Storage (GRS), and Geo-Zone-Redundant Storage (GZRS). This improved my understanding of balancing cost, availability, durability, and disaster recovery requirements.

Role-Based Access Control

Configuring storage permissions so that each department could access only its assigned resources required careful planning.

Resolution

I implemented Azure Role-Based Access Control (RBAC) using Microsoft Entra ID security groups. Instead of assigning permissions directly to individual users, permissions were assigned to departmental groups. This approach simplifies administration, supports scalability, and follows

Project Refinements

After completing the original deployment, I identified several improvements that enhanced the project's quality.

Refinement 1: Strengthening Access Control

Originally, permissions could have been assigned directly to users.

The refined solution assigns permissions to Microsoft Entra ID security groups.

Benefits

- Easier administration, Reduced configuration errors and Better scalability

REFINEMENT 2: APPLYING THE PRINCIPLE OF LEAST PRIVILEGE

Instead of granting broad administrative permissions, each department now receives only the minimum level of access required to perform its responsibilities.

Examples include:

- Sales and Marketing access only to Sales storage, Engineering access only to Engineering storage and Customer Success access only to Customer Success storage

This refinement significantly reduces the organization's attack surface

Refinement 3: Improved Documentation

I expanded the project documentation to include:

- Resource naming conventions, Departmental access assignments with Security configuration steps.
- Deployment decisions and Troubleshooting notes.

Lessons learned

Comprehensive documentation improves maintainability and simplifies future administration.

Refinement 4: Security Validation

Before considering the deployment complete, I added verification procedures to confirm that:

- Authorized users could access their assigned resources.
- Unauthorized users were denied access.
- RBAC assignments functioned correctly.
- Storage permissions matched organizational requirements.

Testing before deployment helps identify security issues early and reduces operational risk.

Ethical Considerations

Cloud administrators are entrusted with protecting sensitive organizational information. This project reinforced several important ethical responsibilities.

Data Privacy

Departmental information should remain accessible only to authorized personnel. Proper RBAC implementation protects confidential business information and supports user privacy.

Ethical Considerations

Confidentiality

Unauthorized access to storage resources could expose financial records, customer information, engineering documents, or proprietary business data. Restricting access protects organizational confidentiality.

Accountability

Every configuration change should be documented and performed responsibly. Administrators must understand the impact of their decisions before implementing changes in production environments.

Integrity

Professional integrity requires correcting security weaknesses immediately rather than ignoring them for convenience or speed. Secure deployments demonstrate responsible stewardship of organizational resources.

IMPACT OF THE REFINEMENTS

The refinements produced several important improvements.

Security

- Reduced unauthorized access, Improved identity management, Stronger permission control, Better protection of organizational data

Reliability

- More consistent deployment process, Reduced administrative errors and Improved resource organization

Scalability

Using Microsoft Entra ID security groups allows the organization to grow without significantly increasing administrative complexity.

Future departments can be added using the same security model.

PROFESSIONAL GROWTH

This project significantly expanded my understanding of:

- Microsoft Azure Administration
- Microsoft Entra ID
- Azure RBAC
- Azure Storage
- Cloud Governance
- Cloud Security
- Identity and Access Management
- Infrastructure Planning

Perhaps most importantly, I learned that cloud administration is not simply about deploying resources—it is about designing secure, reliable, and ethical technology solutions that protect users and organizational data.

LESSONS LEARNED

This project taught me several valuable lessons:

- Careful planning prevents deployment failures.
- Security should be considered from the beginning of every project.
- Documentation is essential for long-term success.
- Testing improves deployment quality.
- Ethical decision-making is just as important as technical expertise.
- Continuous learning is necessary because cloud technologies evolve rapidly.

These lessons will influence how I approach future cloud administration projects throughout my career.

CONCLUSION

Completing this Azure Infrastructure Deployment project has been one of the most valuable learning experiences of my Information Technology education. It allowed me to apply classroom knowledge to practical cloud administration tasks while strengthening my understanding of security, governance, identity management, and ethical responsibility.

By refining the project through improved access control, stronger security validation, better documentation, and adherence to the principle of least privilege, I transformed the initial deployment into a more secure, scalable, and maintainable cloud environment.

This project has prepared me for future roles in cloud administration and technical support by reinforcing the importance of combining technical excellence with integrity, accountability, and service.

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